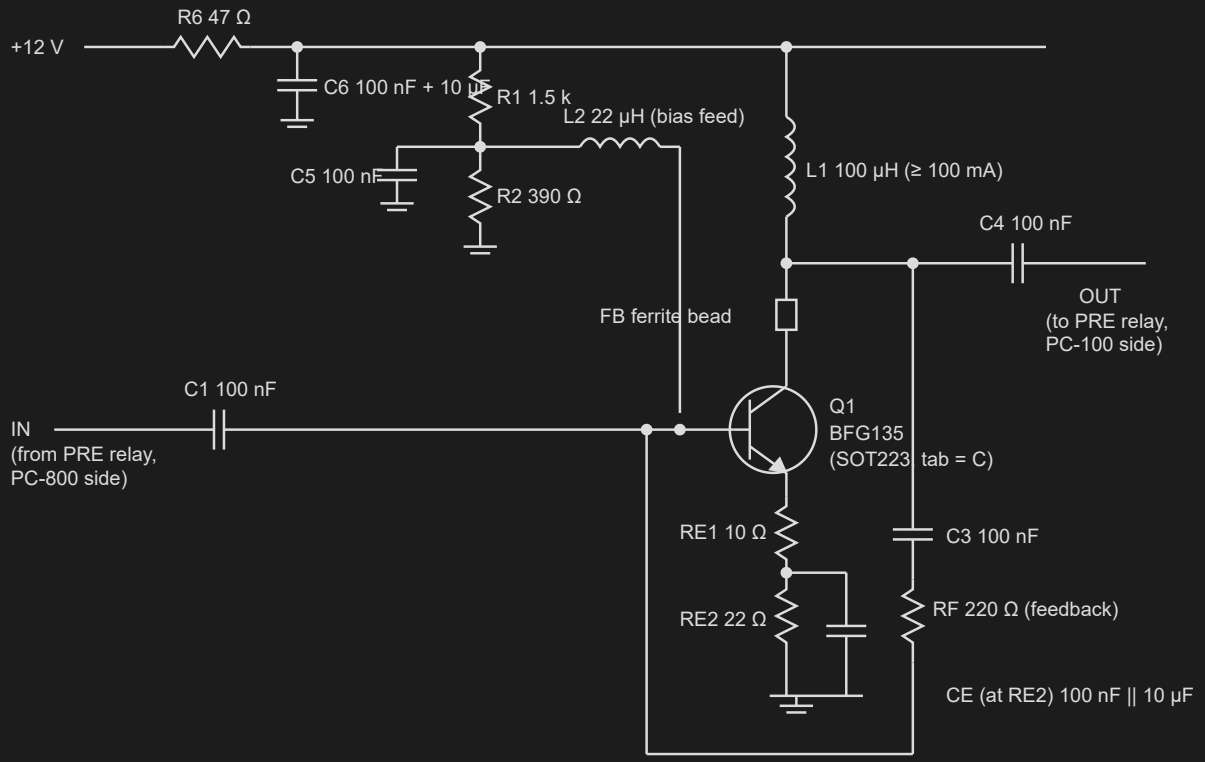
RX preamp — BFG135 broadband feedback amplifier

YT3CT · 2026-10 · proposal · ~10–11 dB, 50 Ω in / out, 3.5–30 MHz, I_C ≈ 50 mA at 12 V · sits on the PRE relay between PC-800 OUT and PC-100 pin 1



Parts

Part	Value	Note
Q1	BFG135	SOT223, collector tab soldered to a copper pad (heat, ~0.4 W)
RF	220 Ω	sets gain / 50 Ω match (RF = Z0 · (1 + gain))
RE1	10 Ω	unbypassed, RF feedback (RE1 = $Z0^2$ / RF)
RE2 / CE	22 Ω / 100 nF ‖ 10 µF	DC stabilisation, bypassed for RF
R1 / R2	1.5 k / 390 Ω	base ≈ 2.4 V → I_C ≈ 50 mA
L1	100 µH	collector choke, rated ≥ 100 mA
L2	22 µH	keeps the divider off the input
R6 / C6	47 Ω / 100 nF + 10 µF	supply filter
C1, C3, C4, C5	100 nF	ceramic, short leads
FB	ferrite bead	on the collector lead — against VHF oscillation

Notes

- Expected: **~10–11 dB** gain, 50 Ω in / out, flat 3.5–30 MHz. More gain: RF 270 Ω + RE1 8.2 Ω (~12 dB).
- Check after building: collector ≈ 9.6 V, emitter ≈ 1.6 V (= 50 mA). Adjust R2 if far off.
- BFG135 has f_T ≈ 7 GHz: build on a ground-plane board, very short leads, all ground returns to the plane.
- Supply: permanent +12 V, or switched on with PRE (PNP from the PRE coil low end) to save ~60 mA.
- On the PRE relay: pole 1 switches the input, pole 2 the output (bypass when off). RX only — never in the TX path.
- Test with the tinySA (sweep through the amp, 50 Ω both ends) or by ear on 10 / 15 m.